

CS3025

Deliverable 4

30/03/2026

Group 9

Members	
Divinefavour Edung	3740935
Peyton Hooper	3719092
Adarsh Budhna	3766711
Isaiah Dakin	3697768

Table of contents:

Section Title	Page(s)
Introduction, P.A.C.T, & System Design	3-10
Final System & UI Design	10-14
Usability Study Plan	14-18
Analyzing Results	19-21
References	21-22

Section 1: Introduction, P.A.C.T, & System Design

1.1: Brief System Overview

The proposed information kiosk system is a theoretical permanent replacement of temporary information booths which pop up around the Head Hall-ITC complex early in the fall and winter semesters. The system is designed to assist visitors, students, and faculty members with navigation and other frequently asked questions regarding the Head Hall-ITC complex.

1.2: System Goals

1. Improve ITC Complex Navigation

The system aims to assist its users in quickly locating relevant classrooms, laboratories, offices, etc. within the ITC wing via a built-in wayfinding feature.

2. Provide a Central Source of Reliable Information

The kiosk system is designed to act as a central information terminal, providing answers to frequently asked questions and other information relevant to computer science.

3. Support the Needs of a Diverse User Pool

The system will support a diverse set of users, such as faculty members, staff, students, and visitors, who may have varying levels of prior knowledge regarding the building layout and computer science at UNB.

4. Provide a Permanent Support Solution

The information kiosk will act as a permanent solution and replacement of temporary information booths in the Head Hall-ITC complex. Additionally, the system will provide ongoing support to users throughout the year.

5. Encourage User Feedback

The system will both encourage and provide opportunities for users to share feedback such as confusion, bugs, and general suggestions. This will allow the system to be iteratively improved to best fit user needs.

1.3: P.A.C.T

1.3.1: People

Head Hall's ITC complex is visited by many people each day. This includes but is not limited to staff, students, faculty, and other general visitors. Every visitor may be seeking information regarding the building or other relevant computer science information; additionally, each visitor will have a varying level of knowledge regarding these subjects. Shown below in Figure 1.0 is a segregation of potential users and their corresponding needs.

Audience		
Primary	Secondary	Tertiary
Students	Visitors	Faculty and staff
Use of the kiosk		
Wayfinding Locating staff and faculty CS-events information Co-op information Accessing library and other services	Wayfinding Contextual information of CS department	Checking event information Verifying contact or office information Verifying locations

Figure 1.0

Additionally, the information kiosk will have a set of base user requirements which must be fulfilled by the system for every user regardless of their degree of prior knowledge or position in Figure 1.0.

1. Focus on Ease of Use & learning

The information kiosk system's UI must be hastily learnable, with an additional focus on ease of use. Findings of prior usability studies show that user interfaces which employ flat navigation and simple layout structures significantly reduce user error rate and cognitive effort (Ekşioğlu et al., 2018; Nielsen, 1994). As such, the kiosk system will employ simple layout structures and flat navigation techniques. For example, Figure 1.1 shows our low fidelity prototype lab page which is located only one UI layer deeper than the home page. Furthermore, this principle applies to all other activity pages, making the overall design consistent with the definition of flat UI design.

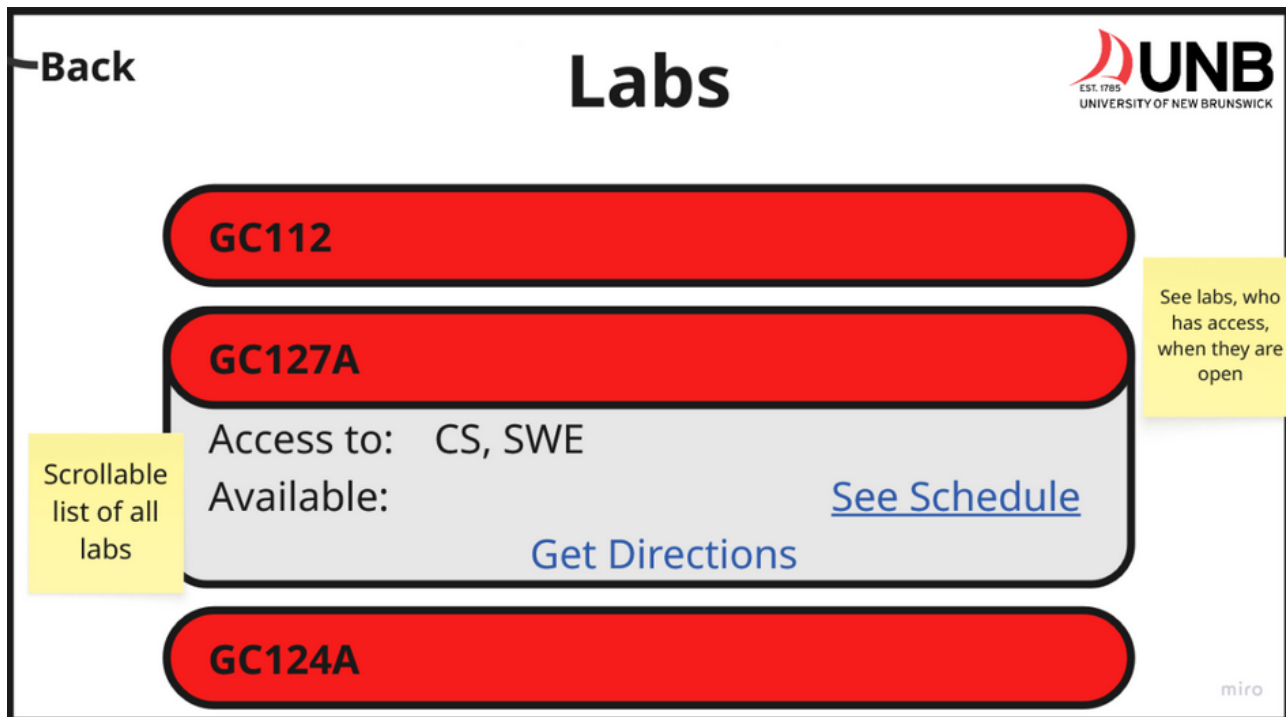


Figure 1.1

2. Support Wayfinding / Pathing

A similar study conducted in 2018 concluded that 75% of new university campus visitors struggled finding specific locations on campus, and that 56% of visitors desired a campus map to assist them in wayfinding (Ekşioğlu et al., 2018). In accordance with these findings, the Head Hall-ITC kiosk system includes a built-in wayfinding feature to assist visitors in locating areas within the ITC complex. Shown below in figure 1.2 is our low fidelity prototype version of the kiosk system's wayfinding feature.

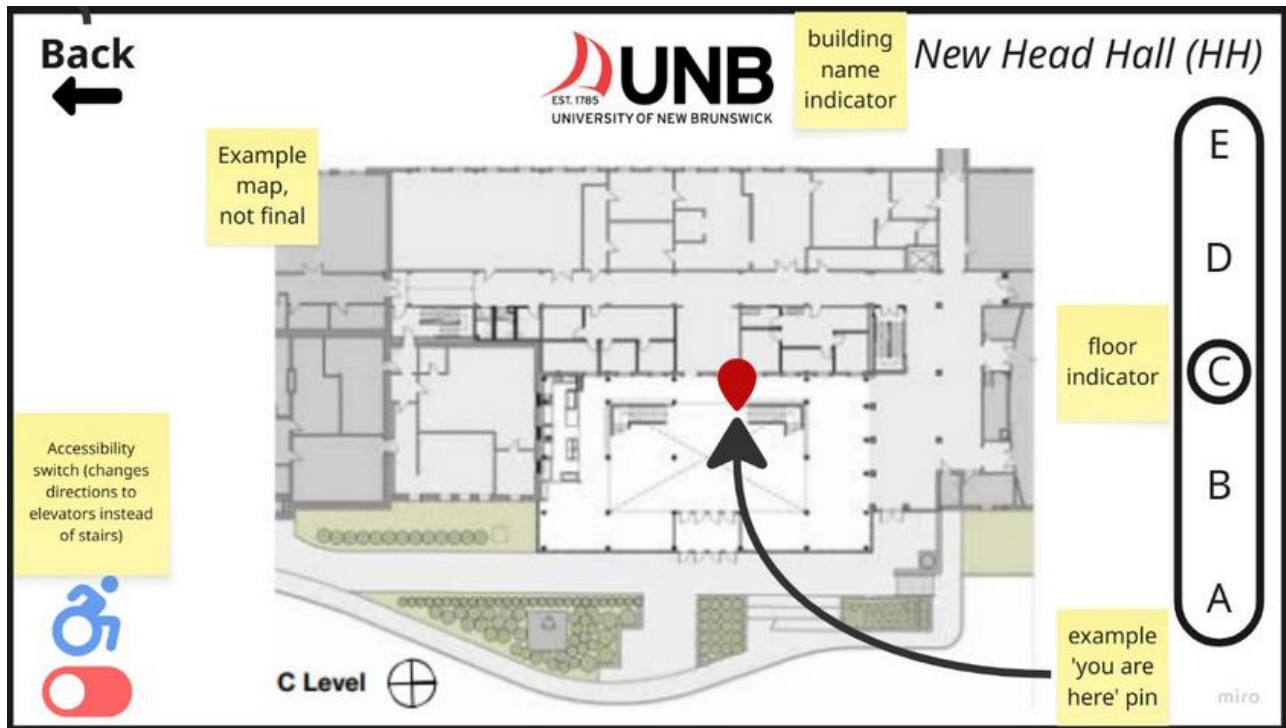


Figure 1.2

3. Provide Reliable & Accurate Information

The information provided by the information kiosk must be up-to-date, accurate, and reliable no matter the time of year. This concept was shown in a prior study, which concluded that users of kiosk systems expect the information provided by the kiosk to be accurate without the involvement or assistance of staff (Ekşioğlu et al., 2018; Litsey et al., 2015).

4. Support High Error Tolerance

As shown by a prior study, users of information kiosks expect the system to have effective and understandable methods of error recovery (Ekşioğlu et al., 2018). The Head Hall-ITC kiosk is designed to reflect these findings.

5. General User Inclusivity

As shown in a similar study, kiosk systems, in their design, must be inclusive of all users; this includes accommodating user's differences in physical, cognitive, and linguistic ability (Ekşioğlu et al., 2018; Newell & Cairns, 1993).

1.3.2: Activities

To best represent the Head Hall-ITC kiosk system's user activities, we first asked a small pool of fellow students and classmates (verbally & electronically) what features they thought would be most useful in the proposed system. As discussed in *Module 6: User Centered Design, PACT Part 4*, involving users who represent our target demographic provides accurate and informed design decisions. To ensure that we gathered informed, accurate, and representative information, we asked students who often reside in Head Hall; specifically, computer science, and engineering students. Shown below in Figure 1.3 is a chart of the most frequently requested features.

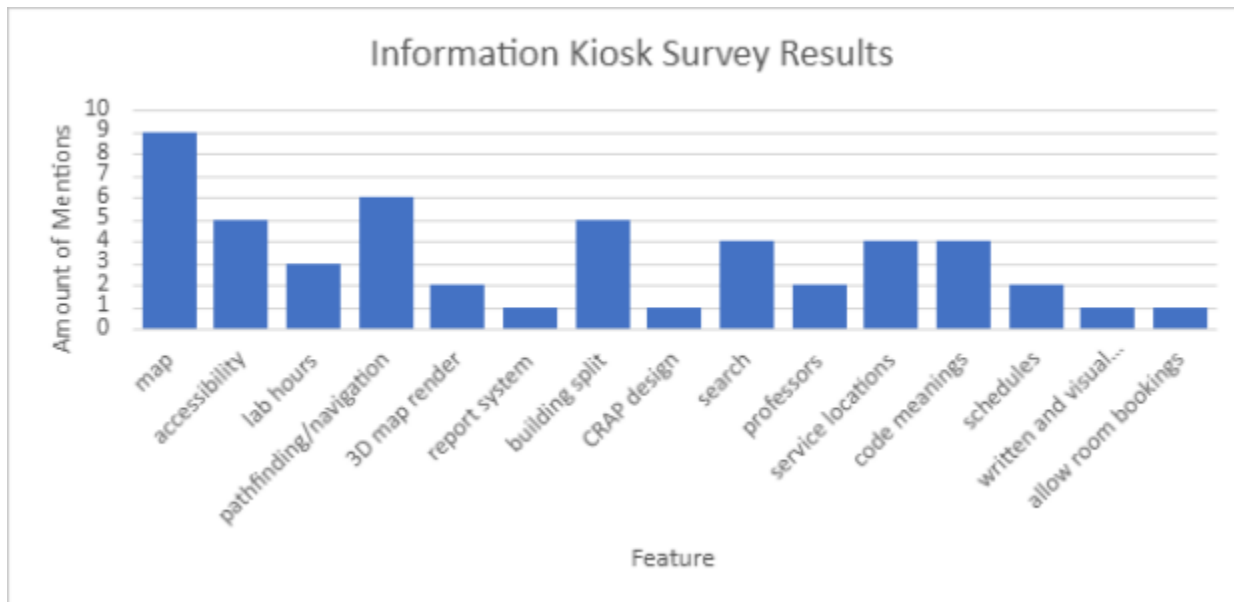


Figure 1.3

As shown by the findings of Figure 1.3, the most requested feature for the system was a map feature, whose goal is to show the general building layout, support wayfinding / pathing features, and support switching between multiple floors individually. Though the map was the most requested feature, respondents commonly proposed other features like accessibility options, lab & classroom information, information regarding professors and services, and more.

Based on the information provided by respondents, we constructed a low fidelity prototype which displayed roughly how we expected to design the real kiosk system. Figure 1.4 shows our low fidelity prototype version of the kiosk system's home menu. For the purposes of familiarity, we used common practices of UI development, CRAP design principles, and HCI usability principles. To make the kiosk system feel familiar to users, we designed its UI in a way in which a user may apply their knowledge from prior computer interactions while interacting

with the kiosk (Module 8: Ten Simple Usability Principles). For example, we placed the search bar along the top, a position which is common for web browsers and apps. Furthermore, since a map was the most requested feature, we made it the single largest UI element and placed it in the center of the page. The CRAP principle of proximity was applied when placing the cluster of activity buttons, and in grouping the buttons on the left and right columns (Lewis University. (2025). Lewisu.edu). For easy visual acquisition of UI elements, we applied the CRAP principle of contrast by highlighting activity buttons in bright red with high visibility white fonts and backgrounds (Lewis University. (2025). Lewisu.edu). Though the color choices happen to be contrasting, they were also chosen to match UNB's official colors, providing an additional layer of familiarity. Finally, CRAP principles of repetition and alignment were used in the appearance and equidistant placement of UI elements on the page (Lewis University. (2025). Lewisu.edu).

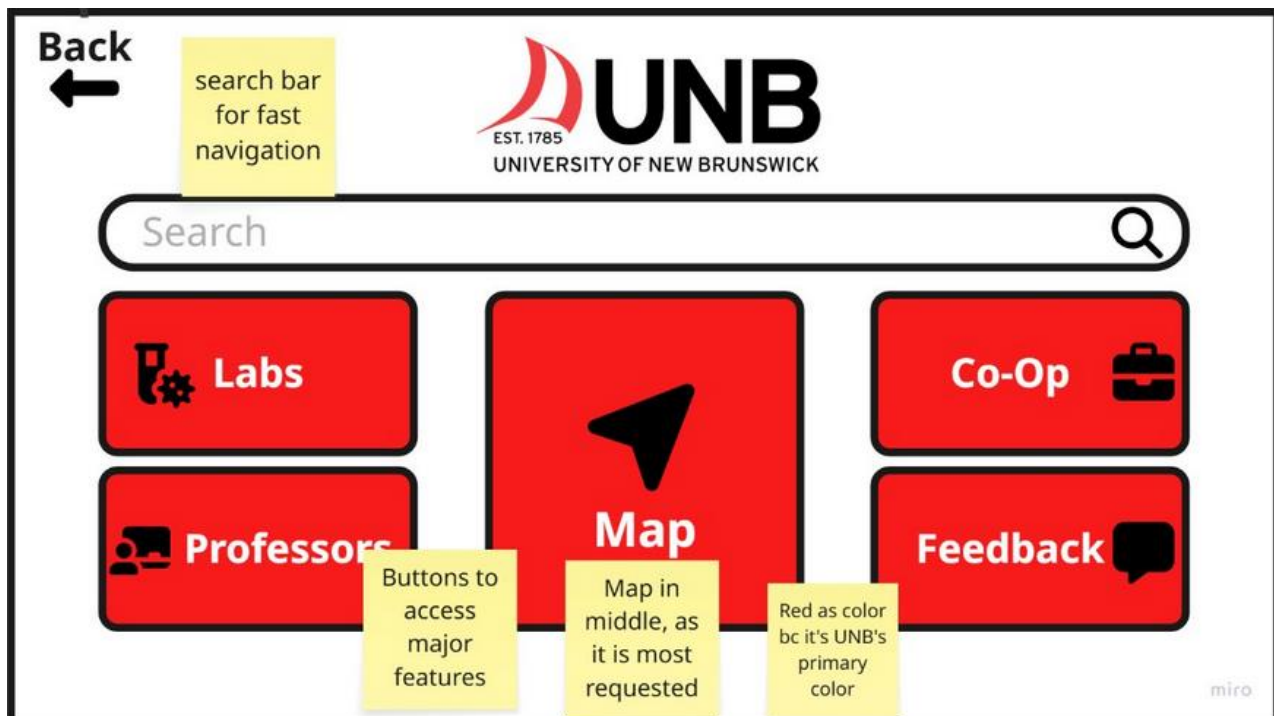


Figure 1.4

1.3.3: Context

The Head Hall-ITC information kiosk, in actual implementation, needs to account for certain contextual considerations such as the public nature of its placement, the target user demographic, and university rules. To address the systems P.A.C.T context, we approached context in three sub-groups similar to *Module 6: User Centered Design, PACT Part 4*: physical, social, and organizational.

The physical context of a system refers to its physical location, and the corresponding constraints applied by that location (Module 6: User Centered Design, PACT Part 4). Perhaps fittingly, the Head Hall-ITC kiosk is intended to be placed in Head Hall's ITC wing. Usability considerations of this location may include loudness, crowdedness, or a generally overwhelming environment. To address these considerations, we planned to implement the following set of rules:

1. The kiosk system should not use audio-based queues or feedback by default.
2. The system should use visual and or haptic feedback by default.
3. The kiosk system should be placed in an area which is public and visible but removed from heavy ITC traffic flow.

As discussed in Section 1.3.1, the primary target demographic of the system is infrequent or first-time visitors of the Head Hall-ITC complex. Given this assertion, we may also assume that users of the system will primarily be novice users who benefit from a supportive environment (Module 6: User Centered Design, PACT Part 4). To address these considerations, we kept novice users in mind, and implemented the following rules:

1. The kiosk should use straight-forward UI design, simple terminology, and low steps per interaction.
2. The system should provide a "help" menu or manual to assist users with common questions.
3. The system should, by default, provide users with feedback via clear visual and or haptic feedback methods.

In consideration of organizational context, the system must adhere to the rules, constraints, and implications of the university. As shown by a similar study, kiosk system users greatly benefit from its ease of use (Ekşioğlu et al., 2018; Nielsen, 1994). In other words, the system ought to be easy to pick up and use to accommodate rushed or otherwise limited windows for interaction, as students often only have a 10-minute window to locate and travel to their classes. To comply with the organizational context of the system, we created the following set of rules:

1. The system's UI should implement large font size and UI elements.
2. The system should have minimal latency.
3. The kiosk system should minimize textual instructions and focus on visual instructions.

1.3.4: Technology

The kiosk system was both planned to be and actually implemented using the web development tool React.js. We made this design choice due to our team's prior experience using React.js, and due to the cross-platform compatibility of web apps. This allowed us to run the kiosk system web-app locally via lab machines and personal computers for the purpose of performing project research but also allowed us to plan for the real-world implementation of the system.

In actuality, the kiosk was planned to run on a regular desktop PC with a touch-screen monitor, or on a large tablet. Due to the unattended nature of information kiosk systems, the kiosk must run in a "kiosk mode", which prevents users from exiting the kiosk web-app and accessing the rest of the operating system. As supported by prior research, the system was planned to include a reset feature, which involves the system automatically returning to the home menu and clearing any fields after a short period of inactivity (Nielsen, 1994; Litsey et al., 2015). Due to the kiosk's touchscreen only nature, the system was planned to accommodate with the following design features:

1. Large & clearly labelled activity buttons and other UI elements.
2. Shallow / flat navigation.
3. Minimal requirement for text-based user input.

These planned design features also conform to human centered design considerations for public touchscreen systems as discussed in *Nielsen, 1994; ISO 9241-210, 2019*.

Because of the system's web-based nature, it would simply require an outlet and stable internet connection for operation. To ensure reliability and minimal latency, the system is planned to be connected to one of UNB's private networks. Though it is out of the scope of this project, our low fidelity plan also included planned caching of some important data locally to support continued kiosk operation through network outages. These features are supported by the findings of a similar study which concluded that public information systems much like the Head Hall-ITC kiosk are recommended to always provide the highest degree of availability possible (Ekşioğlu et al., 2018; Litsey et al., 2015).

Section 2: Final System & UI Design

The final high-fidelity prototype of our system implemented the full feature list and satisfied the system's intended goals as discussed in Section 1. The high-fidelity system prototype has several major organizational differences from the low. However, it still implements all of the planned low-fidelity features and options. Moving forwards, this section will provide visuals and comments for each of the high-fidelity prototype's major features, while providing a short summary of minor features.

2.1: Major Features

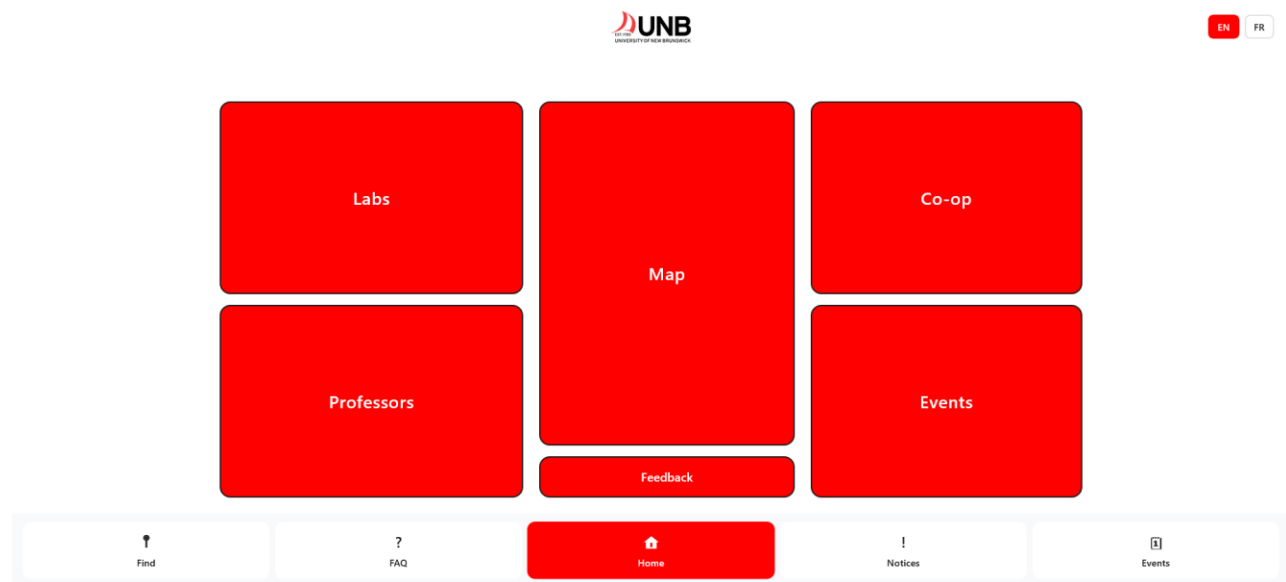


Figure 2.0: Home Menu

Shown above in Figure 2.0 is the high-fidelity kiosk home menu. The activity button grid follows a pattern much like the one seen in our low-fidelity prototype with the addition of the “events” button. This tab was added because of the survey results seen in Figure 1.3, and the low-fidelity prototype’s lack of a “schedule” feature. The purpose of the events tab is to provide kiosk users with a general schedule or list of computer science events and important dates. One major difference which is immediately noticeable is the removal of the “splash screen” for language selection from the low fidelity prototype. We decided that to maintain a lower depth of navigation, it would be optimal to remove the splash screen and instead have a simple language toggle in the top right corner. Furthermore, to decrease cognitive strain, we decided to use the CRAP and usability design principle of proximity to group common quick access features like searching (find), home, and FAQs in a navigation bar along the bottom border.

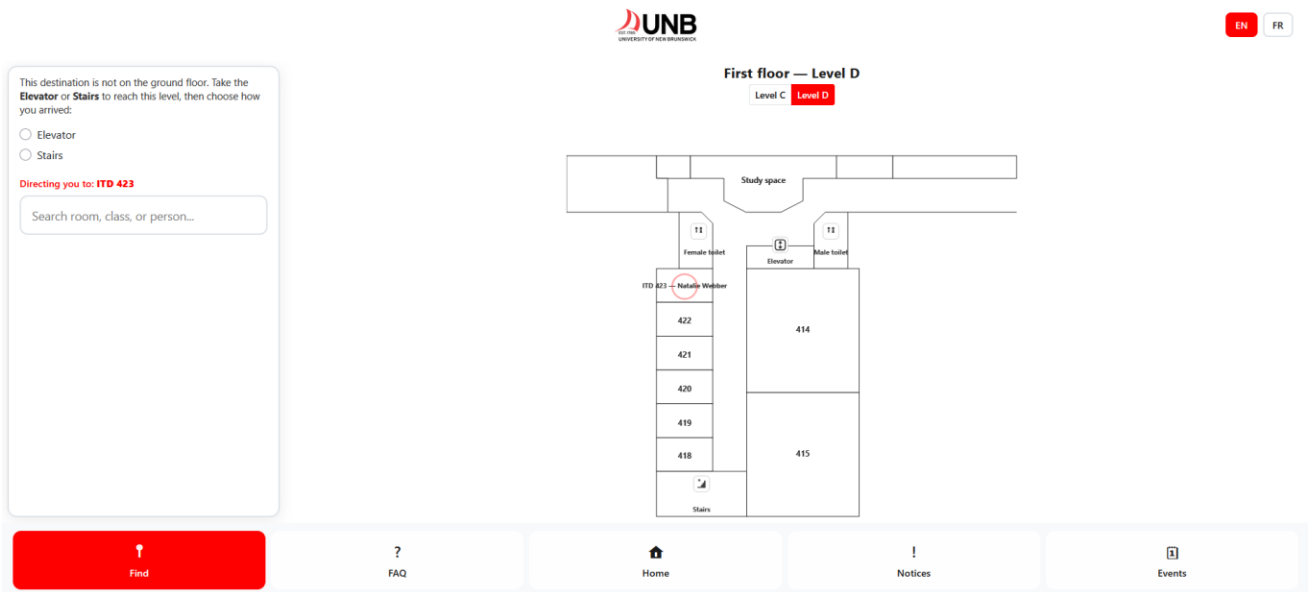


Figure 2.1: Map Page - Navigational Options

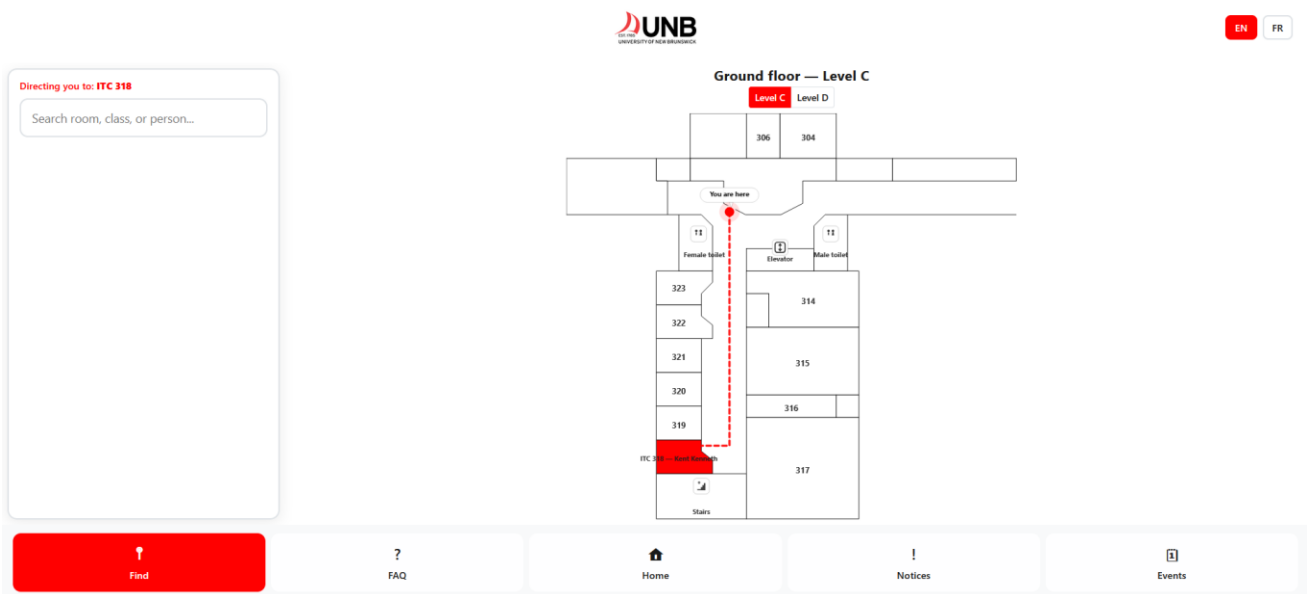


Figure 2.2: Map Page - Wayfinding System

The high-fidelity map feature is shown above in Figures 2.1 & 2.2. Hovering over a specified room shows the room's title or use as seen in Figure 2.1. Clicking on a specific room draws a path from the user's current location to their target destination as seen in Figure 2.2. Additionally, the system accommodates the ITC complex's C and D levels. When a user selects a destination which is not on the ground floor, they are presented with navigational options in the window on the left-hand side of the screen. These options allow the user to path find using either

stairs or elevators, accommodating for differently abled users and their needs. The system then draws a path from the kiosk to the selected option (elevator or stairs). After a fixed short segment of time, the system then automatically switches to the D level and draws a path from the selected option to the target destination. For additional quality of life, the left-hand window also allows the user to search for a specific room code or person, providing users who preemptively know their destination's title with quick access to the system's navigational tools.

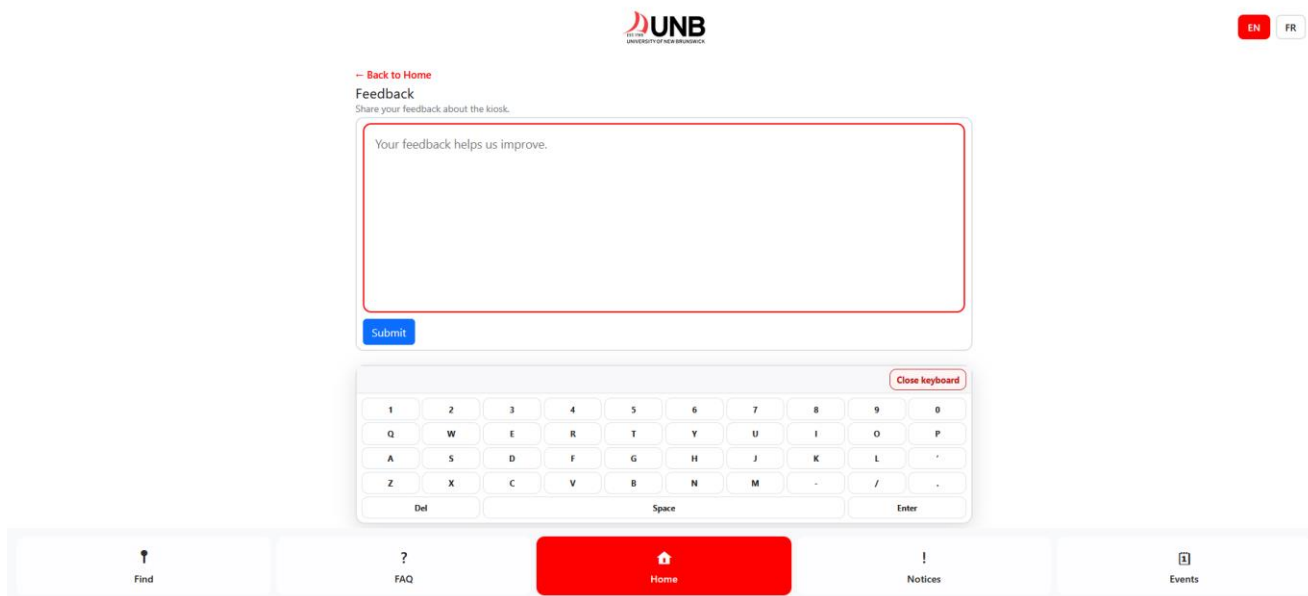


Figure 2.3: Feedback Page

As discussed in Section 1.2, one of our team's planned goals for the system was to encourage and provide opportunities for users to leave feedback. The high-fidelity prototype version of the system achieved this goal by implementing the feedback page shown in Figure 2.3. Upon clicking the "Feedback" button on the kiosk home menu, as seen previously in Figure 2.0, the user is presented with the above screen. For touch screen support, an on-screen keyboard automatically opens allowing the user to type in their feedback. The user then presses the blue "Submit" button, and the system responds with a visual feedback window reassuring the user that their feedback was successfully submitted.

2.2: Minor Features

Minor features of the system are ones which we consider necessary but less important & complicated than the major features discussed in Section 2.1. The minor features in our high-fidelity prototype include the "Professors", "Labs", "Events", "Notices", "FAQ", and "Co-op"

pages which can be accessed from the home menu as seen in Figure 2.0. Each one of these pages provides a list of relevant information regarding the title of the page. For example, the “Labs” page can be seen below in Figure 2.4. The Labs page, similar to all other minor features, provides an organized list of laboratory rooms and lecture halls around the ITC complex. For simplicity’s sake, we kept the information for each minor feature page entry as concise as possible. These features, though simple, are designed to help kiosk users find the information they need as fast as possible when under the pressure of a time crunch.

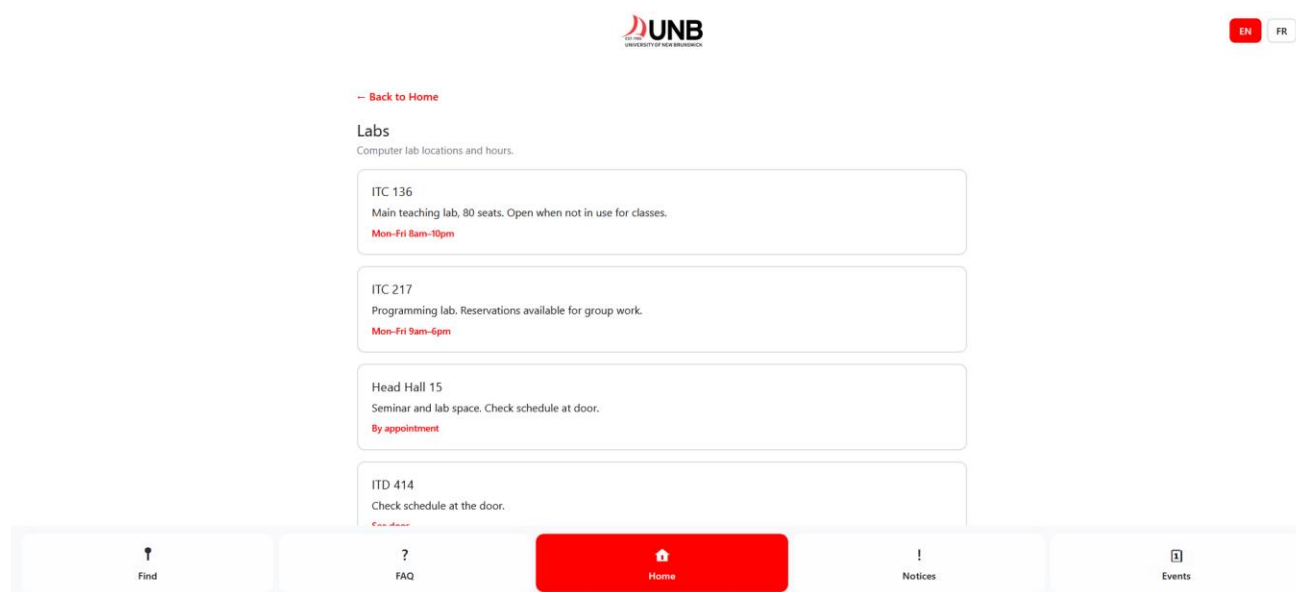


Figure 2.4: Labs Page

Section 3: Usability Study Plan

3.1: Study Goals & Overview

For the usability study of our information kiosk system, we wanted to ensure each feature of our implementation was as quick and easy to learn and to use as possible. For each task given to our participants, a target time was decided to indicate our goals for how quickly we expect users to find the information they are looking for. The target times were chosen to reflect the minimum time between classes that students are given (10 minutes). A leading design goal of our implementation is to reduce the amount of time users spend using our system by presenting the desired information as clearly and as quickly as possible, to allow our target users more time to navigate to and from the information kiosk prior to and following uses.

For example, in Task 4, students are asked to find a navigable route from the information kiosk (next to the service desk on the ground floor) to the office of Natalie Webber (ITD 423)

using the stairs. This task was designed to test the speed at which participants were able to use the navigation feature within our implementation. The target time set for this task was 45 seconds. This was chosen as the length of time we expect our system to be able to give the required information, and for the user to understand the directions. Our study found that the average time taken to complete Task 4 was 34.00125 seconds, more than 10 seconds faster than our target time. This shows that our implementation performed better than expected on average, with none of our participants taking more than the target 45 seconds to complete the task. More information on the target times and task list can be found later in this paper.

Following the completion of the task list, the participants are given a Likert scale questionnaire regarding their experiences with the system. These questions are designed to allow us, the testers, to interpret how the system made participants feel during the study regarding different features. One of the Likert scale questions asks about the “Map” feature, and how easy to learn and to use participants found it to be. This feedback is important, as the “Map” was the most requested feature in our pre-design survey (see Figure 1.3). Our participant responses averaged to a 4.4375, suggesting that participants in our study strongly agreed that the *Map* feature was both easy to use and to learn. Once again, more detail on the Likert scale questionnaire responses will be given later in this paper.

Finally, participants are given time to make any comments regarding their time with the system that the Likert scale questionnaire was unable to cover. These statements will be grouped by similarity, thus allowing common criticisms and feedback to be more easily identified. Using this process, flaws we overlooked in our system, such as the confusion in the overlap between “Notices”, “FAQ”, and “Events”, are more readily apparent if a future iteration of our information kiosk system were to be implemented.

3.2: Usability Factors & Dependent Variables

3.2.1 Learnability and Efficiency

One of the lead design principles of our information kiosk implementation is to minimize the amount of time users spend using our system. To achieve this, we attempt to maximize the affordability of all on-screen components. By making buttons big and contrast the background, they afford interactability, which is the intended outcome. The more affordances our system provides the user, the more quickly users can learn, and the more efficient the users can use the system.

Our system also incorporates several Gestalt Principles (Module 11: Gestalt Principles - Part 2), such as Proximity and Similarity, implemented in the groupings of buttons on the “Home” screen and along the quick bar. The implementation of these principles helps our system to be easier for participants to navigate and use.

The Usability Principles (Module 8: Ten Simple Usability Principles) are also incorporated into our information kiosk system. In a system designed to be fast, users should be able to acquire information in a way that makes the most sense to them. By introducing Flexibility into our system, users can find, for example, Professor Natalie Webber's office number via the "Map" or "Professor" screens. Furthermore, Recoverability is important in our system, as users are likely to need to recover quickly in order to complete tasks quickly. Most screens have a "Back" button to return the user to the "Home" screen. Alternatively, interacting with the "Home" button in the quick bar will also allow users to recover to the "Home" screen, combining both Recoverability and Flexibility.

The inclusion of these Gestalt and Usability Principles is to maximize learnability and efficiency, to ensure users spend reduced time with the system, and still achieve the results they intended.

3.2.2 Accessibility

In order to make our information kiosk as user-friendly as possible several accessibility features were implemented. Firstly, as New Brunswick is a bilingual province, both provincial languages, English and French, are available in our system via a toggle in the upper right corner, accessible on every screen. Those with extra mobility needs are accounted for, by allowing our map's pathfinding to reroute to the elevator instead of the stairs. This is done by selecting 'Elevator' or 'Stairs' when attempting to pathfind to a location on the D level (upper floor) on the map. Those with visual impairment were also kept in consideration and was a supporting reason for the high contrast colors for our buttons and background. This contrast should allow those with any kind of colorblindness to still use our system effectively.

3.2.3 Dependent Variables

The dependent variables in our study are the times in which participants are able to complete the given tasks from the task list and the responses given in both the Likert scale questionnaire and open feedback sections. Each test was taken in the same environment, using the same system, with the same tasks, in the same order. These controlled variables ensure that the variables being tested are impacted as little as possible from other factors (nuisance variables). While participants' varying levels of familiarity with the ITC Wing of Head Hall does have an impact on the results of the tests, this was mitigated by choosing participants who have somewhat similar time spent in and around the ITC Wing, that being primarily computer science students from upper years (see Participant Recruitment below).

3.3: Usability Study

Since all participants are given as close to the same conditions as possible, there are no independent variables. This is the case for many usability studies. What we are measuring is not

the comparison between several alterations in a system, but rather the effectiveness of the system itself. This leads us to rely more on reaching goals (see Section 3.1) and interpreting user feedback. By analyzing the responses to the questionnaire and open response sections, insight into the effectiveness of our information kiosk implementation can be obtained, removing the requirement for altering independent variables for our study.

3.4: Tasks & Participant Grouping

Prior to the start of the test, participants were given a verbal overview of both the system and the process of the study. The system overview was given so that the participants had a general understanding of the feature set prior to their attempts to complete the given tasks. The explanation of the process of the study was to allow participants to know what they would be asked to do before they began.

Each participant would be given the same task list, consisting of twelve tasks, all achievable using our information kiosk implementation. The tasks were designed to cover the breadth of our system's features, ensuring participants interacted with the variety of functionalities that our system is designed to cover in day-to-day usage. Each task had a target answer that the testers would look for, and the task would be deemed 'complete' once the participant provided this answer. Here is the list of tasks, in order, that participants were asked to complete:

- Who's office is located in ITC 320?
- How many computer labs are in the ITC Wing?
- What is Natalie Webber's office number?
- How do you get from the kiosk to Natalie Webber's office using the stairs?
- How do you get from the kiosk to Natalie Webber's office using the elevator?
- On what floor level is Wei Song's office located?
- When is the last date to add/drop courses without academic penalty this term?
- What CS event occurred on January 25?
- What is Patricia Evans' role, in French?
- Leave feedback to report a broken elevator.
- What are the Wi-Fi network options presented to visitors?
- At what time does Head Hall open?

Following the completion of the task list, participants were asked to complete a seven-part Likert scale questionnaire about their experience with the system. Their responses to the questionnaire are designed to get feedback about their experiences and feelings while using the system. Here is a complete list of the Likert scale questions:

- I found that the kiosk system was easy to learn and use.
- Using the kiosk system was a very frustrating experience for me.

- I found the kiosk system helpful when finding room information, co-op information, and other academic information.
- I found that the kiosk's Map feature was easy to learn and use.
- In my experience, the kiosk system does not make me feel stressed and is very pleasant to use.
- I believe that this kiosk system would be helpful for first-time or infrequent visitors of the ITC complex.
- I think that UNB should implement a kiosk system like this one in the Head Hall building.

Lastly, participants were given a chance to provide additional feedback and comments that the Likert scale questionnaire did not cover. This was to allow participants to give more specific insights into their perspective of the system, as well as highlight areas they'd like to see changed and/or improved in the system.

3.5: Participant Recruitment

Participants were recruited from the faculties of both computer science and engineering. Participants were chosen from these groups, as they are frequent visitors to the Head Hall-ITC complex, as mentioned in section 1.3.2. Additionally, our system implementation focused more specifically on the ITC wing of Head Hall, therefore computer science students made up a vast majority of our participants. As the time constraints for this study were limited, we were able to gather only upper year students from the faculties, which means our system's effectiveness for new and infrequent visitors to the ITC Wing, such as first year students, is only predicted by responses from the questionnaire and open feedback.

Participants were asked (verbally and electronically) for their voluntary participation in our study. On the dates of the study periods, participants were asked to sign a consent form prior to the start of their participation. This ensured that participants were aware of both the process of the study and the usage of data collected from their participation.

3.6: Measurements & Result Analysis

After the explanations mentioned in section 3.4, participants were read the first task in the list, and then the tester would start an external stopwatch via their smartphone. Once the target answer for each task was presented, the test taker would press the 'lap' button on the stopwatch, measuring the time taken for that task to be completed, while continuing the running time, to measure a total completion time for the entire task list. Upon the completion of the final task, the stopwatch would be stopped, and the lap times and total time would be recorded.

The Likert scale questionnaire was read aloud to participants, and their response for each statement was recorded. Following the questionnaire, participants were asked to give any additional feedback or comments about the system. All feedback given was recorded and added to the participants' results.

Section 4: Analyzing Results

4.1: Task Results Analysis

As discussed in Section 3, participants were given a sequence of 12 tasks using the kiosk system. For each of the 12 tasks, we estimated a time which was reasonable per task and the maximum allotment of time in class transitional periods. Participants' times per task were recorded and then averaged as shown in Figure 4.1. The results show that participants, on average, completed the tasks in a time less than the estimated time for the majority (9/12) of the tasks. Additionally, 14/16 participants completed all 12 tasks in a time less than the estimated total time frame. Furthermore, out of 192 total tasks completed by participants throughout the study, only two DNFs were recorded, meaning that participants were able to successfully complete 99% of tasks across the board.

Generally speaking, the statistical results we derived from our study support the hypothesis that the proposed kiosk system would allow for navigation of the ITC complex to a degree which is higher than navigation without the system.

Shortcomings in the averages for tasks 7, 9, and 12 highlight an underlying issue with the kiosk's "Notices" page and language toggle option. These shortcomings were primarily caused by participant confusion between information held by the "Notices" and "FAQ" pages, and by confusion when attempting to locate the language toggle. These issues were common subjects of the participant's open-ended responses, which will be discussed further in Section 4.3.

Task	Task 1	Task 2	Task 3	Task 4	Task 5	Task 6	Task 7	Task 8	Task 9	Task 10	Task 11	Task 12	Total (min)	Total (sec)
AVG	8.863125	17.19875	12.64125	34.00125	21.043125	15.55375	22.34875	10.303125	20.061875	21.1465875	14.65875	29.738125	3m48.111875	228.111875
Target	15	20	15	45	30	20	20	15	20	30	15	20	4m25s	265
Hit Ratio	15 for 16	11 for 16	12 for 16	16 for 16	13 for 16	13 for 16	8 for 16	14 for 16	12 for 16	14 for 16	11 for 16	5 for 16	N/A	N/A

AV														
G <														
Tar														
get	yes	yes	yes	yes	yes	yes	no	yes	no	yes	yes	no	yes	yes

Figure 4.1

4.2: Exit Survey Response Analysis

After participants completed the 12 tasks, they were asked to complete a short seven question Likert Scale exit survey regarding their experience with the kiosk system. Shown below in figure 4.2 are the averages of participant responses per question. For every question below, except for question 2, higher value responses (strongly agree) are better. In other words, the exit survey shows that participants had a very positive experience using the system across the board.

Question 6 was worded as follows, “I believe that this kiosk system would be helpful for first-time or infrequent visitors of the ITC complex”. As shown in Figure 4.2, question 6 had a 4.8125 average response value, meaning that 96.25% of participants strongly agreed that a kiosk system like ours would be helpful to our target demographic. This analysis further supports our studies' hypothesis by directly connecting participant responses to it.

Question	Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Question 7
Average Response	4.5625	1.4375	4.5625	4.4375	4.6875	4.8125	4.75

Figure 4.2

4.3: Open-Ended Response Analysis

The open-ended response section of the study was added to allow for more precise feedback that the Likert scale questionnaire does not provide. After collecting comments from our sixteen participants, we are able to identify common themes that help show areas where the system could be improved. Of our sixteen participants, thirteen had additional feedback to give about the system, with some providing substantially more than others.

The most common feedback received was the confusion around the purpose of “Notices”. Six participants commented that the “Notices” provided information that they expected to be under the “FAQ” section, with one participant saying *‘I intrinsically thought the information would be*

in “Events”. This confusion is easily apparent in task 12, which asks participants what time Head Hall opens in the morning. Only five of our sixteen participants were able to retrieve the correct answer of 7am in the target time of 20 seconds. By observation alone, testers noted that nearly all participants went to both the “FAQ” and “Events” pages prior to checking the “Notices” page for this information. This design oversight resulted in the average completion time for task 12 to be nearly 50% higher than the target time (29.738s vs 20s). Of the six who expressed problems with “Notices”, four of them suggested combining the information presented in “Notices”, “FAQ”, and “Events” (in some combination). One participant proposed doing away with the “Notices” option in favor of a “Calendar” option, which could show both upcoming events and important dates and times in one place, while dividing the information from “Notices” between “FAQ” and “Events” where applicable.

Another theme that appeared in multiple participant comments was that of “Feedback”. Three participants stated the term ‘feedback’ was too vague for its usages. One participant suggested “Issues/Feedback” as a replacement name, which covers the use as a reporting feature for building issues, like a broken elevator or water fountain. Another user proposed labels or severity levels in the “Feedback” feature, helping indicate priority of reports. These suggestions for more clarity within the “Feedback” feature could help reduce confusion surrounding the intended use cases for the feature.

Other themes of note include introducing more iconography into the system (2 participants) to allow for better accessibility for those with issues reading, more interconnectivity between the “Professors” and “Map” screens (2 participants), which would allow for faster interactions between the two, and sort/filter options for “Labs”, “Professors”, and “Events” screens (2 participants). These features would increase usability of our system, as they would increase both Flexibility and Accessibility within our system. Some suggested features (such as more “Professors”/“Map” connection) were planned originally, but were cut from the scope due to time constraints. If future iterations of our information kiosk system were to be made, these features would be high on the priority list of things to add.

4.4: Conclusion

Our team set out to create an information kiosk implementation for the Head Hall complex, to aid all visitors of the building in navigation and information collection as quickly and as efficiently as possible. Our test results show that, while flawed, our system can quickly provide the necessary information to navigate to classes and offices, to find dates for faculty events, and does so while being easy to use and learn. A majority of feedback from the Likert scale questionnaire indicates a strong positive experience from our study participants when using our system for the given tasks. The feedback received from the open response section shows there is plenty of room for our system to grow, but we have built a strong foundation within our information kiosk implementation. Beyond our system, our study finds that even students in the

upper years of their education, who are more familiar with the building layout, would like to see the implementation of a system, like the one we proposed, throughout the Head Hall complex.

References

- Ekşioğlu, M. (2016). *User experience design of a prototype kiosk: A case for the İstanbul public transportation system*. *International Journal of Human–Computer Interaction*, 32(10), 802–813.
- Ekşioğlu, M., Güler, H., Terzi, F., Yıldırım, H. S., & Yücel, B. (2018). User experience design for a prototype campus information kiosk. In *Proceedings of the 2nd International Conference on Computer-Human Interaction Research and Applications (CHIRA 2018)* (pp. 176–182). SCITEPRESS.
- Litsey, R., Hidalgo, S., Daniel, K., Barnett, J., Kim, A., Jones, S., & Ketner, K. (2015). Interactive kiosk at the Texas Tech University Libraries. *Journal of Access Services*, 12(1–2), 31–41.
- Newell, A. F., & Cairns, A. Y. (1993). *Designing for extraordinary users*. Cambridge University Press.
- Nielsen, J. (1994). *Usability engineering*. Academic Press.
- Webber, N. (2026). *Module 6: User Centered Design, PACT Part 4*. <https://lms.unb.ca/d2l/le/content/278344/viewContent/3207333/View?ou=278344>
- Webber, N. (2026). *Module 8: Ten Sample Usability Principles*.
<https://lms.unb.ca/d2l/le/content/278344/viewContent/3207333/View?ou=278344>
- Webber, N. (2026). *Module 11: Gestalt Principles - Part 2*.
<https://lms.unb.ca/d2l/le/content/278344/viewContent/3207333/View?ou=278344>
- Lewis University. (2025). *Lewisu.edu*.
<https://www.lewisu.edu/writingcenter/pdf/crap-resource-revised-pub>
- International Organization for Standardization. (2019). *ISO 9241-210:2019 Ergonomics of human-system interaction: Human-centred design for interactive systems*.
- International Organization for Standardization. (2008). *ISO 9241-171:2008 Ergonomics of human-system interaction: Guidance on software accessibility*. <https://www.iso.org>
- Dix et al., Human–Computer Interaction Dix, A., Finlay, J., Abowd, G. D., & Beale, R. (2004). *Human–Computer Interaction* (3rd ed.). Pearson Education.

